SHREEL PATEL

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PROFESSIONAL SUMMARY

Mechatronics Engineer with a strong electrical engineering focus, delivering schematic design, manufacturable PCB development, and low-voltage system architecture for real hardware systems. Experienced in power budgeting and efficiency driven design trade-offs to meet strict performance requirements. Seeking a hands-on electrical and control systems role where reliability, efficiency, and disciplined engineering practice are critical.

PROFESSIONAL EXPERIENCE

Team Arrow Racing Association **November 2025 – Current**
Mechatronics Engineer (Electrical Team)

- Contributing to the design and refinement of electrical schematics using Altium Designer, supporting power distribution, serial communication integration (CAN) and subsystem integration.
- Assisting with component selection, interface definition and schematic review to improve electrical robustness and system reliability.
- Producing structured engineering documentation, including schematic annotations and test observations to support traceability, design validation and iterative system integration.
- Designed and optimised LED PCBs to meet specified candela targets while minimising power usage, applying hand calculations and component selection to balance efficiency and performance

Greg Chappell Cricket Centre **August 2023 – Current**
Part-time Retail Sales Assistant

- Communicated clearly with customers and suppliers using Microsoft outlook, ensuring accurate and reliable handling of enquiries, warranty claims and follow ups in line with company policies.
- Used Microsoft Excel and Microsoft Teams to track warranty cases and stock movement creating an organised backlog for effective issue resolution.
- Worked in a collaborative fast paced work environment, coordinating tasks and priorities to maintain consistent customer service quality during trading hours.

EDUCATION

Queensland University of Technology ***2023 – Current***
Bachelor of Engineering- Mechatronics & Computer and Software Systems (Honours)
Current GPA: 6.217/7

Expected Graduation: November 2026

PROJECTS

Embedded System for an Electrical Vehicle

- Designed and implemented a 3-phase BLDC motor subsystem in C using FreeRTOS, integrating hall effect sensor feedback with closed loop PID control to achieve a <2% steady state RPM error.
- Integrated sensing, system interface and motor control subsystems using RTOS kernel primitives such as semaphores and mutexes to allow for deterministic and safe interthread communication.
- Implemented and validated a hard real-time monotonic scheduling approach to minimize input latency, while ensuring seamless motor and sensor interaction.

- Verified control system stability and timing behaviour through iterative testing and tuning, assessing response under varying conditions.

Autonomous pick and pack warehouse Robot

- Designed a robust 3D printed chassis, modelled through inventor using iterative prototyping to maximise the durability of the robot while minimising the weight.
- Designed and integrated a differential drive system through MATLAB based gear ratio analysis to meet speed and torque requirements.
- Implemented closed loop PID speed control with differential tracked wheels, enabling stable translational and rotational motion while maintaining traction.
- Collaborated in a multidisciplinary team to integrate the electrical mechanical and software subsystems harmoniously to achieve zero functional errors during testing.

TECHNICAL SKILLS

Simulation/Design: LTspice, AutoCAD, Inventor, Altium Designer

Programming: Python, C/C++, C#, MATLAB

Tools: GitHub, Linux, VS Code, Jupyter, FreeRTOS, Altium 365, Microsoft Office

REFERENCES

On Request